G. BURCH FISHER

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Updated - 10/9/2026

EDUCATION

2013 Ph.D. in Earth Science, University of California at Santa Barbara
Advisors - Douglas Burbank and Bodo Bookhagen

2007 M.S. in Earth Science, Dartmouth College
Advisor - Frank Magilligan

2004 B.A. cum laude with *honors* in Geology, Middlebury College
Studied Abroad at La Universidad de La Serena, Chile (February to June 2002)
Advisor - Peter Ryan

Postdoctoral Advisors
University of Pittsburgh - Eitan Shelef
University of Texas at Austin - Independent Researcher
University of California at Santa Barbara - Douglas Burbank and David Valentine

PROFESSIONAL EXPERIENCE

2024–Present	Associate Research Scientist, University of Maryland Center for Environmental Science
2018–present	Environmental/Geospatial Data Consultant
2022–2024	Researcher, University of Maryland Center for Environmental Science
2021–2022	Researcher - National Socio-Environmental Synthesis Center (SESYNC)
2017–2024	Affiliate Researcher, Earth Research Institute, University of California at Santa Barbara
2016–2017	Postdoctoral Associate, University of Pittsburgh
2014–2017	Assistant Researcher II, Earth Research Institute, University of California at Santa Barbara
2014–2016	Jackson Distinguished Postdoctoral Fellow, University of Texas at Austin
2014	Assistant Specialist III, Marine Science Institute, University of California at Santa Barbara
2013–2014	Assistant Specialist III, Earth Research Institute, University of California at Santa Barbara
2007–2013	PhD Candidate, University of California at Santa Barbara
2009–2012	NASA Earth and Space Science Fellow
2005–2007	MS Candidate, Dartmouth College
2004–2005	Research/Teaching Assistant, Middlebury College
Summer 2004	Research Assistant, University of California at Santa Cruz
Summer 2003	Intern, Kenai Watershed Forum/Alaska Conservation Foundation
Winter 2003	Intern, USGS-BRD Kilauea Field Station, Volcanoes National Park, HI
Summer 2002	Intern, Güirá Oga (Bird and Mammal Recuperation Center), Puerto Iguazu, Argentina (Spanish)
Spring 2002	Intern, Pukara (AIDS Organization), Coquimbo, Chile (Spanish)

AWARDS & HONORS

2014-2016	Jackson Distinguished Postdoctoral Fellowship, Jackson School of Geosciences, UT Austin
2013	G.K. Gilbert Award for Best Graduate Student Presentation, UCSB Department of Earth Science
2009-2012	NASA Earth and Space Science Fellowship (NESSF)
2009	Reds Wolman Graduate Student Research Award, AAG - Geomorphology Specialty Group
2008	Richard and Eleanor Migues Field Research Award, UCSB Department of Earth Science

2007	NSF Graduate Research Fellowship Honorable Mention
2006	Geological Society of America Outstanding Student Research Award (Top 17 grants out of 550)
2005	Dartmouth Presidential Fellowship Award
2005-2007	Dartmouth Fellowship (Tuition and Stipend)
2005	Best Graduate Student Presentation, 42nd Annual Mtg. of the Clay Minerals Society
2004	3rd place talk at the Vermont Geological Society Meeting

WORKSHOPS & CONTINUED EDUCATION

AI Unlocked: Empowering Higher Education through Research and Discovery Workshop, Denver, CO, June 2-3, 2026

AI Unlocked: Empowering Higher Education through Research and Discovery Workshop, Denver, CO, April 2-3, 2025

Leveraging Artificial Intelligence and Machine Learning to Advance Chesapeake Bay Research and Management: A review of status, challenges, and opportunities, Smithsonian Environmental Research Center, MD, February 24-25, 2025

iHARP/UMBC High Performance GPU Computing Bootcamp, Baltimore, MD, November 12, 2024

DataCamp Data Scientist with Python Career Track (Completed Summer 2018, ~100 hours of coursework)

CUAHSI NASA Remote Sensing Hydrology Workshop, Cambridge, MA, April 4-6, 2017 (GRACE and SMAP)

Geospatial Data Analysis in R, UT Summer Statistics Institute, Austin, TX, May 23-26, 2016

ExxonMobil Deepwater Reservoir Modeling Field Course, La Jolla, CA, June 2011

Google Earth: Visualizing the Possibilities for Geoscience Education & Research, GSA Penrose Conference, Googleplex, Mountain View, California, January 2011

Riegl Terrestrial LiDAR Training, Orlando, Florida, June 2008

Gulf of Maine River Barrier Removal Monitoring Workshop, Orono, Maine, June 2006

SYNERGISTIC ACTIVITIES

Co-instructor (along with Kayo Ogilby) of a two-week, intensive GIS learning experience entitled “Hot Springs Hunter: A Geothermal Viability Study for the Town of Paonia”, Colorado Rocky Mountain School, Carbondale, CO, February 7-18, 2022.

Co-convener (with William Amidon) of the *Burbank Research Symposium* honoring the career of Professor Douglas Burbank, Santa Barbara, CA, May 19-21, 2017.

Senior scientist on the SEEPS ’15 (Studies of the Evolution and Ecology of Petroleum Systems) research cruise in the Gulf of Mexico (June 2015 on the R/V Atlantis) in charge of shipboard multibeam sonar bathymetry mapping.

Development of open source software (*ChanGeom*) and manuals to quantify channel planform geometry from freely available aerial and satellite imagery. (www.burchfisher.com/data.html)

Visiting researcher at the University of Wollongong, Australia (January 2012)

Visiting graduate student at the University of Potsdam, Germany (November 2010)

Leader/organizer of Dartmouth Earth Sciences Dept. field trip entitled “Structural Evolution of the Champlain Valley”

Assisted in organizing and producing the 2005 meeting of the Clay Minerals Society in Burlington, VT

TEACHING EXPERIENCE

Workshops/Tutorials/Webinars

NAIRR Pilot, CU RMACC YouTube Webinar/Tutorial (Invited with Xiaoxu Guo). Led “Introduction to AI-Assisted Coding: Prompt Engineering with LLMs in Google Colab”, **LINK**

AI Unlocked: Empowering Higher Education through Research and Discovery Workshop, Denver, CO, June 2-3, 2026. Led/co-led (along with Xiaoxu Guo and Roy Prouty) two workshop tutorials including: 1) *Introduction to AI-Assisted Coding: Prompt Engineering with LLMs in Google Colab* and 2) *Introduction to Predictive Modeling: A Hands-On Machine Learning Tutorial Using Observational Environmental Data* tutorials. We (along with Sai Vikas Amareni) also organized a BoF (birds of a feather) discussion titled

"*Toward an AI Skills Inventory: Effectively Assessing Advanced Cyberinfrastructure Upskilling Interventions at Research Institutions*"

UMCES/UMBC Hands On Research Computing and Science Collaborations Workshop, Annapolis, MD, April 2-3, 2026 (Four separate workshop sessions focused on using LLMs to code, AI/ML pipelines, interactive data dashboards, and image processing)

UMCES/UMBC CGC-SCIPE Introduction to NSF ACCESS Infrastructure and Jetstream2, Zoom, July 16, 2025

UMCES/UMBC Hands On Research Computing and Science Collaborations Workshop, IMET, Baltimore, MD, May 28, 2025 (Focus on using LLMs to code and AI/ML pipelines in Google Colab)

Instructor of Record, UC Santa Barbara

Mountains, Boots, and Backpacks: Field Study of the High Sierra (EARS 6 w/ D. Burbank, J. Harvey)(Sum 2012)
Geological Applications of GIS (EARS 176) (Spring 2012)
Geology and the Environment (EARS 1) (Summer 2009)

Graduate Teaching Assistant, UC Santa Barbara

Quantitative Geomorphology (Geography 137 – B. Bookhagen) (Fall 2008)
Global Warming - Science and Society (Geology 130 – D. Lea) (Spring 2008, 2009)
*Guest lectured first three lectures of Geology 130 for D. Lea (Spring 2009)
Plate Tectonics (Geology 157 – D. Wilson) (Winter 2008)
Field Studies in Geological Science (Geology 18 – T. Atwater) (Fall 2007)
Mountains, Boots, and Backpacks: Field Study of the High Sierra (Geology 6 – D. Burbank) (Fall 2007)

Graduate Teaching Assistant, Dartmouth College

Guest Lecturer for Earth Surface Processes and Landforms (Geography 33 – F. Magilligan) (Spring 2007)
Introduction to Earth Science (Earth Science 1 – J. Aronson & J. Kaste) (Spring & Fall 2006)

Teaching Assistant, Middlebury College

Oceanography (Geology 161 – T. Manley) (Fall 2004 & 2005)
Marine Geology (Geology 342 – P. Manley) (Fall 2004)
Geographic Information Systems (Geography 320 – R. Churchill) (Fall 2003 & Spring 2004)
Vertebrate Natural History (Biology 302 – S. Trombulak) (Fall 2002)

GRANTS AWARDED

2025-2026 **NSF ACCESS Discover Request (BIO250269) (400,000 Credits) - Co-PI with Xin Chen, PI Matt Fitzpatrick (UMCES)**
A near real-time biodiversity mapping, monitoring, and risk assessment system for the Central Appalachians

2025-2026 **NSF ACCESS Discover Request (EES250056) (1,500,000 Credits) - Co-PI with Xiaoxu Guo, PI Victoria Coles (UMCES)**
HPC Testbed for Research and Training across UMCES

2024-2025 **National Fish and Wildlife Foundation #0403.23.079338 to American Bird Conservancy - Sub-award ($77,000 of $685,723.43), PI Jeff Larkin (IUP)**
Restoring Dynamic Forest Structure for Priority Birds in the Delaware River Watershed: Phase 7 (PA): : Advancing synergy between public and private forest lands in PA

2017 **Global TanDEM-X DEM Grant - PI, Co-PI Shelef - (DEM data awarded)**
Landscape response to large-scale drainage capture in northwest Argentina: Evidence from high-resolution TanDEM-X topography

2014-2016 **Jackson Distinguished Postdoctoral Fellowship, University of Texas at Austin - ($140,000)**

2009-2012 **NASA Earth and Space Science Fellowship - Co-PI with Bookhagen - ($90,000)**

ASTER-Derived River Widths and Their Spatial Implications for Erosion in the Tectonically Active Himalaya

2011 **UCSB Science and Engineering Research Grant - ($2000)**
Quantifying Soil Production in the Tectonically Active Himalaya

2009 **GSA Graduate Student Research Grant - ($1570)**
Quantification of Climate-Erosion Coupling in the Himalaya - A Continuing Education Experience

2009 **ExxonMobil Global Geoscience Grant - ($5000)**

2008 **ExxonMobil Geoscience Grant - ($7500)**
Spatial and Temporal Quantification of Climate-Erosion Linkages in the Himalaya

2007 **Preston Cloud Memorial Travel Grant, UCSB Department of Earth Science - ($300)**

2006 **GSA Graduate Student Research Grant - ($2700)**
Bring Back the Salmon! Understanding the Role of LWD in Salmon Recovery in Downeast Maine

2006 **Dartmouth College Research Grant - ($2000)**
Understanding the Relationship between Large Woody Debris, Sediment Transport, and Channel Complexity in Downeast Maine Salmon Streams

2006 **Clay Minerals Society International Student Travel Grant - ($1000)**

2006 **Dartmouth College Earth Sciences Dept. Conference Travel Grant - ($750)**

2006 **GSA Conference Travel Grant - ($50)**

2004 **Middlebury College Conference Travel Grant - ($550)**

PEER-REVIEWED PUBLICATIONS

† Undergraduate student

29. Koleck, R., D.J. McNeil, L. Chronister, **G. Fisher**, C. Fiss, J. Goldman, L. Gray, T. Hodges, J. Kitzes, S. Lapp, H. Larkin, J.T. Larkin, R. Tyl, J.L. Larkin (2026), Using passive acoustic monitoring and LiDAR to conduct a statewide assessment of ruffed grouse (Bonasa umbellus) occurrence in Pennsylvania, ***Journal of Avian Biology***, e03587, doi:10.1002/jav.03587.

28. Fisher, G.B., A.J. Elmore, M.C. Fitzpatrick, D.J. McNeil, J.W. Atkins, and J.L. Larkin (2024), Mapping recent timber harvest activity in a temperate forest using single date airborne LiDAR surveys and machine learning: lessons for conservation planning, ***GIScience & Remote Sensing,*** 16, 1, 2379198, doi:10.1080/15481603.2024.2379198.

27. Fuoco, K.B., D.J. McNeil, C.J. Fiss, C.I. Bocetti, **G.B. Fisher**, and J.L. Larkin (2024), Breeding season space use and habitat selection by Blue-winged Warblers in managed shrublands, ***Avian Conservation and Ecology***, 19, 1, doi:10.5751/ACE-02587-190106.

26. McNeil, D.J., **G. Fisher**, C.J. Fiss, A.J. Elmore, M.C. Fitzpatrick, J.W. Atkins, J. Cohen, and J.L. Larkin (2023), Using aerial LiDAR to assess regional availability of potential habitat for a conservation dependent forest bird, ***Forest Ecology and Management***, 540, 121002, doi:10.1016/j.foreco.2023.121002.

25. Fisher, G.B., †L.V. Luna, W.H. Amidon, D.W. Burbank, B. de Boer, L.B. Stap, B. Bookhagen, V. Godard, M.E. Oskin, R.N. Alonso, E. Tuenter, and L.J. Lourens (2023), Milankovitch-paced erosion in the southern Central Andes, ***Nature Communications,*** 14, 424, doi:10.1038/s41467-023-36022-0.
**Nature Communications Editor's Highlight*

24. Utz, R., S. Bidlack, **B. Fisher**, and S. Kaushal (2022), Urbanization drives geographically heterogeneous freshwater salinization in the northeastern United States, ***Journal of Environmental Quality***, 1-14, doi: 10.1002/jeq2.20379.

23. Leach, L.L., **G.B. Fisher**, and J.M. Gross (2022), Nitroreductase/metronidazole-mediated ablation and a MATLAB platform (RpEGEN) for studying regeneration of the zebrafish retinal pigment epithelium, ***Journal of Visualized Experiments,*** 181, e63658, doi:10.3791/63658.

22. Bryant, W.L., R. Camilli, **G.B. Fisher**, E.B. Overton, C.M. Reddy, D. Reible, R.F. Swarthout, and D.L. Valentine (2020), Harnessing a decade of data to inform future decisions: Insights into hydrocarbon release at Taylor Energy's Mississippi Canyon Block 20 (MC20) site, ***Marine Pollution Bulletin,*** 155, doi:10.1016/j.marpolbul.2020.111056.

21. Scherler, D., R.A. DiBiase, **G.B. Fisher**, and J-P. Avouac (2017), Testing monsoonal controls on bedrock river incision in the Himalaya and Eastern Tibet with a stochastic-threshold stream power model, ***Journal of Geophysical Research - Earth Surface,*** 122, 1389-1429, doi:10.1002/2016JF004011.
**JGR - Earth Surface Editor's Highlight*

20. Amidon, W.H., **G.B. Fisher**, D.W. Burbank, P.L. Ciccioli, R.N. Alonso, †A. Gorin, †P. Silverhart, and †M.S. Christoffersen (2017), Mio-Pliocene aridity in the south-central Andes associated with Southern Hemisphere cold periods, ***Proceedings of the National Academy of Sciences,*** 114, 6474-6479, doi:10.1073/pnas.1700327114.

19. Amidon, W.H., †L.V. Luna, **G.B. Fisher**, D.W. Burbank, A.R.C. Kylander-Clark, and R. Alonso (2017), Provenance and tectonic implications of Orán Group foreland basin sediments, Río Iruya canyon, NW Argentina (23°S), ***Basin Research***, 29, 96-112, doi:10.1111/bre.12139.

18. Bagby, S.C., C.M. Reddy, C. Aeppli, **G.B. Fisher**, and D.L. Valentine (2017), Persistence and biodegradation of oil at the ocean floor following Deepwater Horizon, ***Proceedings of the National Academy of Sciences,*** 114, E9-E18, doi:10.1073/pnas.1610110114.

17. Valentine, D.L., **G.B. Fisher**, O. Pizarro, C.L. Kaiser, D. Yoerger, J.A. Breier, and J. Tarn (2016), Autonomous marine robotic technology reveals an expansive benthic bacterial community relevant to regional nitrogen biogeochemistry, ***Environmental Science and Technology***, 50, 11057-11065, doi:10.1021/acs.est.6b03584.

16. Amidon, W.H., P.L. Ciccioli, S.A. Marenssi, C.O. Limarino, **G.B. Fisher**, D.W. Burbank, and A.R.C. Kylander-Clark (2016), U-Pb ages of detrital and volcanic zircons of the Toro Negro Formation, northwestern Argentina: age, provenance and sedimentation rates, ***Journal of South American Earth Sciences***, 70, 237-250, doi: 10.1016/j.jsames.2016.05.013.

15. Harrison, L.R., T. Dunne, and **G.B. Fisher** (2015), Hydraulic and geomorphic processes in an overbank flood along a meandering, gravel-bed river: implications for chute formation, ***Earth Surface Processes and Landforms***, 40, 1239-1253, doi:10.1002/esp.3717.

14. Valentine, D.L., **G.B. Fisher**, S.C. Bagby, R.K. Nelson, C.M. Reddy, S.P. Sylva, and M.A. Woo (2014), Fallout plume of submerged oil from Deepwater Horizon, ***Proceedings of the National Academy of Sciences***, 111, 15906-15911, doi:10.1073/pnas.1414873111.

13. Godard, V., D.L. Bourlès, F. Spinabella, D.W. Burbank, B. Bookhagen, **G.B. Fisher**, A. Moulin, and L. Léanni (2014), Dominance of tectonics over climate in Himalayan denudation, ***Geology***, 42, 243-246, doi:10.1130/G35342.1.

12. Kaste, J.M., F.J. Magilligan, C.E. Renshaw, **G.B. Fisher**, and W.B. Dade (2014), Seasonal controls on meteoric ^{7}Be in coarse-grained river channels, ***Hydrological Processes***, 28, 2738-2748, doi:10.1002/hyp.9800.

11. Amos, C.B., S.J. Brownlee, D.H. Rood, **G.B. Fisher**, R. Bürgmann, P.R. Renne, and A.S. Jayko (2013), Chronology of tectonic, geomorphic, and volcanic interactions and the tempo of fault slip near Little Lake, California, ***Geological Society of America Bulletin,*** 125, 1187-1202, doi:10.1130/B30803.1.

10. **Fisher, G.B.**, B. Bookhagen, and C.B. Amos (2013), Channel planform geometry and slopes from freely available high-spatial resolution imagery and DEM fusion: Implications for channel width scalings, erosion proxies, and fluvial signatures in tectonically active landscapes, ***Geomorphology,*** 194, 46-56, doi:10.1016/j.geomorph.2013.04.011.

9. Godard, V., G.E. Tucker, **G.B. Fisher**, D.W. Burbank, and B. Bookhagen (2013), Frequency-dependent landscape response to climatic forcing, ***Geophysical Research Letters***, 40, 859-863, doi:10.1002/grl.50253.

8. Amos, C.B., A.T. Lutz, A.S. Jayko, S.A. Mahan, **G.B. Fisher**, and J.R. Unruh (2013), Refining the southern extent of the 1872 Owens Valley earthquake rupture through paleoseismic investigations in the Haiwee area, southeastern California, ***Bulletin of the Seismological Society of America***, 103, 1022-1037, doi: 10.1785/0120120024.

7. **Fisher, G.B.**, C.B. Amos, B. Bookhagen, D.W. Burbank, and V. Godard (2012), Channel widths, landslides, faults, and beyond: The new world order of high-spatial resolution Google Earth imagery in the study of earth surface processes, in *Google Earth and Virtual Visualizations in Geoscience Education and Research*, edited by S.J. Whitmeyer, D.G. De Paor, J. Bailey, and T. Ornduff, ***Geological Society of America Special Paper 492***, 1-22, doi:10.1130/2012.2492(01).

6. Godard, V., D.W. Burbank, D.L. Bourlès, B. Bookhagen, R. Braucher, and **G.B. Fisher** (2012), Impact of glacial erosion on the evolution of ^{10}Be concentration in fluvial sediments of the Marsyandi catchment, central Nepal, ***Journal of Geophysical Research - Earth Surface,*** 117, F03013, doi:10.1029/2011JF002230.

5. **Fisher, G.B.**, F.J. Magilligan, J.M. Kaste, and K.H. Nislow (2010), Constraining the timescales of sediment sequestration associated with large woody debris using cosmogenic beryllium-7, ***Journal of Geophysical Research - Earth Surface,*** 115, F01013, doi:10.1029/2009JF001352.

4. Magilligan, F.J., P.B. Goldstein, **G.B. Fisher**, B.C. Bostick, and R.B. Manners (2008), Late Quaternary hydroclimatology of a hyper-arid Andean watershed: Climate change, floods, and hydrologic responses to the El Nino-Southern Oscillation in the Atacama Desert, ***Geomorphology***, 101, 14-32, doi:10.1016/j.geomorph.2008.05.025

3. Magilligan, F.J., K.H. Nislow, **G.B. Fisher**, J. Wright, G. Mackey, and M. Laser (2008), The geomorphic function and characteristics of large woody debris in low gradient rivers, coastal Maine, USA, ***Geomorphology,*** 97, 467-482, doi:10.1016/j.geomorph.2007.08.016

2. **Fisher, G.B.**, and P.C. Ryan (2006), The smectite to disordered kaolinite transition in a tropical terrace chronosequence, Pacific coast, Costa Rica., ***Clays and Clay Minerals,*** 54, 571-586, doi:10.1346/CCMN.2006.0540504.

1. Manley, T.O., P.L. Manley, and **G.B. Fisher** (2005), Bathymetry of Lake Champlain, Copyrighted by Middlebury College. (Published Bathymetric Map with Explanatory Text and Maps).

TECHNICAL REPORTS

Amos, C.B., A.T. Lutz, A.S. Jayko, S. Mahan, **G.B. Fisher**, and J.R. Unruh (2012), Refining the southern extent of the 1872 Owens Valley earthquake rupture – Paleoseismic investigations at Sage Flat and Haiwee Meadows, California, Final Technical Report to the U.S. Geological Survey National Earthquake Hazards Reduction Program, Award Number G09AP00133

INVITED PRESENTATIONS, INTERVIEWS, PODCASTS, & WEBINARS

2026 **NAIRR Pilot, CU RMACC YouTube Webinar/Tutorial (Co-led by Xiaoxu Guo)**
"Introduction to AI-Assisted Coding: Prompt Engineering with LLMs in Google Colab", **LINK**

2026 **Moving the Needle Podcast, Faculty Center for Teaching and Learning, University of Maryland, Baltimore (Invited Guest Expert)**
"*AI, Unscripted: Sustainable AI – Balancing Innovation with Environmental Responsibility*", **LINK**

2025 **Department of Atmospheric, Oceanic, and Earth Sciences, George Mason University, Fairfax, VA**
Evidence for orbital pacing of erosion from the southern Central Andes during the Pliocene

2024 **Appalachian Laboratory, University of Maryland Center for Environmental Science, Frostburg, MD**
UMCES Computing and Data Professional Seminar

2016 **Department of Geological Sciences, University of Indiana, Bloomington, IN**
Erosional CLUE: Was it Milankovitch in the Andes with a monsoon?

Department of Geological Sciences, University of Indiana, Bloomington, IN
Deciphering form, process, and consequence in transitional landscapes

De Ford Lecture Jackson School of Geosciences, University of Texas, Austin, TX
Erosional CLUE: Was it Milankovitch in the Andes with a monsoon?

2015 **Bureau of Economic Geology, University of Texas, Austin, TX**
The fate and persistence of Deepwater Horizon oil

Bureau of Economic Geology, University of Texas, Austin, TX
Pliocene drivers of erosion in the Andes

Department of Geology and Planetary Science, University of Pittsburgh, Pittsburgh, PA
Orogenic CLUE: Who is responsible for the erosion of mountain belts and when?

2014 **Earth Research Institute, University of California, Santa Barbara, CA**
From googly to Google-eyed: Explicit channel widths from high-spatial resolution Google Earth imagery

2013 **Department of Earth Science, University of California, Santa Barbara, CA**
Deciphering landscape drivers and signatures in tectonically active landscapes using Google Earth, remote sensing, and cosmogenic nuclides

2012 **Climate Research Group, Department of Geography, University of California, Santa Barbara, CA**
From googly to Google-eyed: Hillslope response to high magnitude climatic and tectonic events in Haiti

School of Earth and Environmental Sciences, University of Wollongong, Wollongong, Australia
From googly to Google-eyed: Using high-spatial resolution Google Earth imagery to quantify earth surface processes

2011 **ExxonMobil Exploration Company, Houston, TX**
Channel widths, landslides, faults, and beyond: The new world order of high-spatial resolution Google Earth imagery

2010 **Aspen Center for Environmental Studies** (Presented by Kayo Ogilby)
Himalayan erosion - A continuing education experience

ExxonMobil Upstream Research Company, Houston, TX
Climate-tectonic linkages in the Himalaya: Implications for landscape evolution in tectonically active orogenic systems

PROFESSIONAL AFFILIATIONS

American Geophysical Union (2006 - Present)
Geological Society of America (2004 - 2022)
Association of American Geographers - Geomorphology Specialty Group (2008 - 2016)
Clay Minerals Society (2004 - 2013)

SERVICE

UMCES Institution

2026- Generative AI Guidance Committee

UMCES Appalachian Laboratory

2025- Sustainability Committee
2025-2026 Diversity, Equity, and Inclusion Group Member
2025-2026 Employee Onboarding Committee

UC Santa Barbara - Earth Research Institute

2016 Merit Committee for Promotion of Dr. Matt Rioux

UC Santa Barbara - Department of Earth Science

2012-2013 Speakers Club Committee
2010-2012 G.K. Gilbert Award Committee
2007-2009 Website Development Committee
2007-2009 Educational Outreach Committee

MENTORING

Ph.D. Students

2029 Committee member for Jeff Larkin's Ph.D thesis on using LiDAR to inform forest and wildlife management. Primary advisor - Malcolm Itter in the Environmental Conservation Department at the University of Massachusetts Amherst.

M.S. Students

2021 Committee member for Samantha Bidlack's M.S. thesis on freshwater salinization trends in the eastern United States. Primary advisor - Ryan Utz at the Falk School of Sustainability and Environment at Chatham University.

2018 Co-advised Justin Tran's M.S. thesis at UCSB on the morphology of hydrocarbon induced topography (with David Valentine). Justin was funded through the Science, Mathematics, and Research for Transformation (SMART) Scholarship for Service Program from the Department of Defense.

Undergraduate Students

2026 Mentored Fort Lewis College undergraduate intern Stephen Conklin (class of 2026) working on remote sensing projects integrating Harmonized Landsat Sentinel data and airborne LiDAR in Pennsylvania and South Africa

2016 Mentored and funded undergraduate research assistant Michael Christoffersen (class of 2019) working in the UT Paleomagnetics Laboratory

2014 Two weeks field work in Argentina with Middlebury College undergraduates Perri Silverhart and Drew Gorin

2013 One week of field work along the Yakima River with Western Washington undergraduate Skyler Sorsby

2013 Two weeks field work in Argentina with Middlebury College undergraduate Sebastian Schell

2013 Co-mentored Lisa Luna's undergraduate thesis at Middlebury College on paleo-erosion rates from the Oran section, northwest Argentina (Principal advisor - William Amidon)

2012 Three weeks field work in Argentina with Middlebury College undergraduate Lisa Luna

2011 Mentored Josh Noble's GIS term project investigating tectonic rates from topographic metrics along the Goriganga River in the Indian Himalaya.

High School

2022 Senior project mentor for Aden Berry from the Colorado Rocky Mountain School

2009 Funded, organized, and supervised a continuing education field research opportunity for a high school geology professor in the Himalaya (Kayo Ogilby from the Colorado Rocky Mountain School)

2009 Senior project mentor for Elli Mckinley from the Colorado Rocky Mountain School

JOURNAL REFEREE

Nature Geoscience, Remote Sensing of Environment, Communications Earth & Environment, Environmental Science and Technology, Geophysical Research Letters, Earth Surface Dynamics, Journal of Geophysical Research - Earth Surface, Earth Surface Processes and Landforms, Geomorphology, Journal of Quaternary Science, Remote Sensing, Basin Research, npj Natural Hazards, Earth and Space Science, Sedimentary Geology, Sensors, Deep-Sea Research Part II, Scientia Agricola

FIELD EXPERIENCE

Fluvial Geomorphology – Himalaya, southwestern U.S., northeastern U.S., Peru, Alaska, & Costa Rica
Quaternary Climate Change – Andes, Himalaya, southwestern U.S., Peru, Costa Rica, & Hawaii
Human Impacts on Natural Systems – Southwestern U.S., northeastern U.S., Alaska, & Hawaii
Soil Weathering & Geochemistry – Himalaya, Costa Rica, & Northeastern U.S.
Tectonics & Geomorphology – Andes, Himalaya, southwestern U.S., Peru, Costa Rica
Hydrocarbon Dispersal, Persistence, and Ecology – Gulf of Mexico, California coast

FIELD, LABORATORY, & COMPUTATIONAL TECHNIQUES

Field Work – Terrestrial LIDAR, surveying (handheld GPS, total station, laser range finding, and traditional methods), OSL, CRN, paleomagnetism, and U-Th/He sampling, sedimentology, stratigraphy, and structural/bedrock mapping.

Spatial – Remote sensing (Google Earth Engine, ENVI), GIS (ESRI, QGIS, GRASS, GMT, GDAL), TopoToolbox, River Tools, MicroStation (CAD program for LIDAR), lidR (LiDAR package in R), CloudCompare, and MB-Systems for multibeam sonar bathymetry

Numerical – Matlab, Python (numpy, scipy, pandas, rasterio, geopandas, scikit-learn, xgboost, NetworkX, etc), Unix, R and experience with SQL and Julia

Geochemical – Fallout (FRNs) and cosmogenic radionuclides (CRNs), U-series dating of soils, ICAPS major and trace element analyses, ICP-MS dating of ash and detrital zircons, and limited H and O isotope geochemistry.

Mineralogical – Magnetochronology, quantitative mineralogy (QXRD), XRD, SEM, and petrography.

PRESENTATIONS & ABSTRACTS

† Undergraduate student

40. Chen, H., A.J. Elmore, **G.B. Fisher**, B.E. McNeil, and T. Qiu (2026), Beyond phenological timing: species- and trait-mediated variation in green-up and senescence rates across temperate deciduous trees, *Ecological Society of America (ESA) Annual Meeting.*

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